

# Predicting open system quantum trajectories with deep learning

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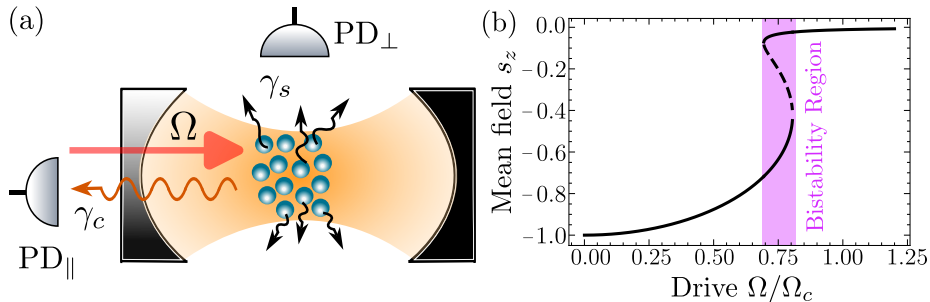
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1 Proposal

2 Fixed  $N$  and  $\Omega$ : Generating more trajectories

# Physics (short): Driven-dissipative cavity QED



(a) System under study (b) Mean-field solution and bistability region

Main equation – Master equation

$$\dot{\rho} = -i[\mathcal{H}, \rho] + \frac{\gamma_c}{2} \mathcal{D}[\hat{S}_-]\rho + \frac{\gamma_s}{2} \sum_n \mathcal{D}[\hat{\sigma}_n]\rho, \quad \mathcal{H} = 2\Omega\hat{S}_x \quad (1)$$

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## How to solve the master equation? Answer: MCWF method

Generally: Solve for density matrix:  $4^N \times 4^N$  elements,  $N$  is the number of qubits.  
Instead: Probabilistic unraveling of the master equation into quantum trajectories

$$|\psi(t + dt)\rangle = \begin{cases} \frac{\hat{S}_- |\psi(t)\rangle}{\sqrt{p_-}} & \text{with probability } p_- = \gamma_c dt \langle \psi(t) | \hat{S}_+ \hat{S}_- | \psi(t) \rangle \\ \frac{\hat{\sigma}_n |\psi(t)\rangle}{\sqrt{p_n}} & \text{with probability } p_n = \gamma_s dt \langle \psi(t) | \hat{\sigma}_n^\dagger \hat{\sigma}_n | \psi(t) \rangle \\ \frac{e^{-i\mathcal{H}_{NH}dt} |\psi(t)\rangle}{\sqrt{p_0}} & \text{with probability } p_0 = 1 - p_- - \sum_n p_n \end{cases} \quad (2)$$

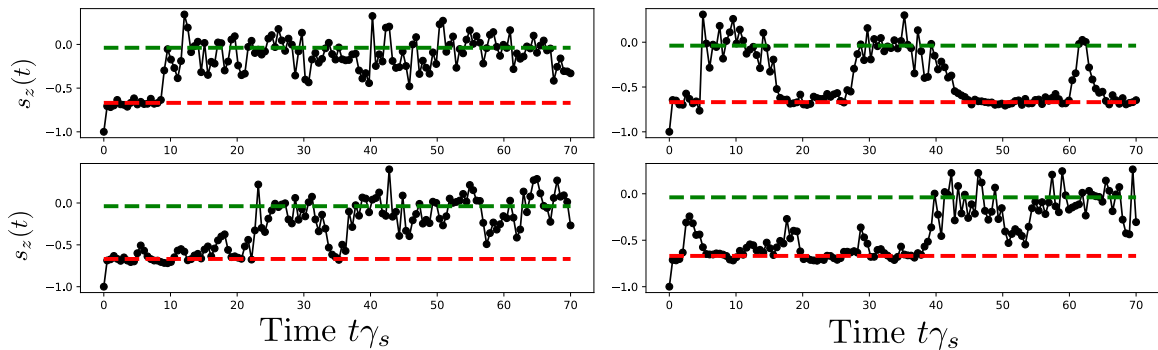
✓ Saves memory:  $2^N \times 2^N$  elements

✗ Loses Time: Need to average over many trajectories to get the density matrix.

In our work we found that a bistability is seen in a single trajectory, but the bistability is lost in the density matrix average.

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## How typical trajectory looks like (in bistability case)

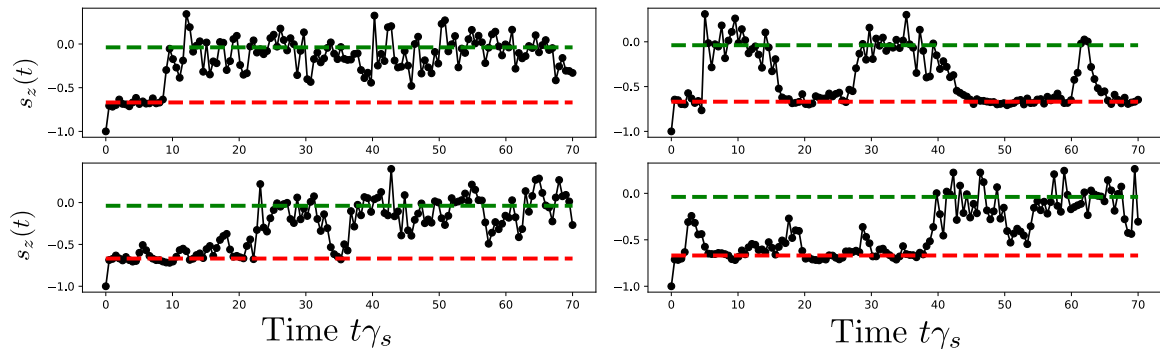


4 examples of quantum trajectories  $N = 18$  (best I can get),  $\Omega = 0.73\Omega_c$  (bistability region),  $\gamma_c = 10\gamma_s$ . Red and Green lines are the mean-field solutions. Generated with QuTiP (python library for quantum simulations).

Note: average of  $N_{tr} \approx 100$  trajectories gives the density matrix solution, which is a smooth curve (no bistability).

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# Problem statement

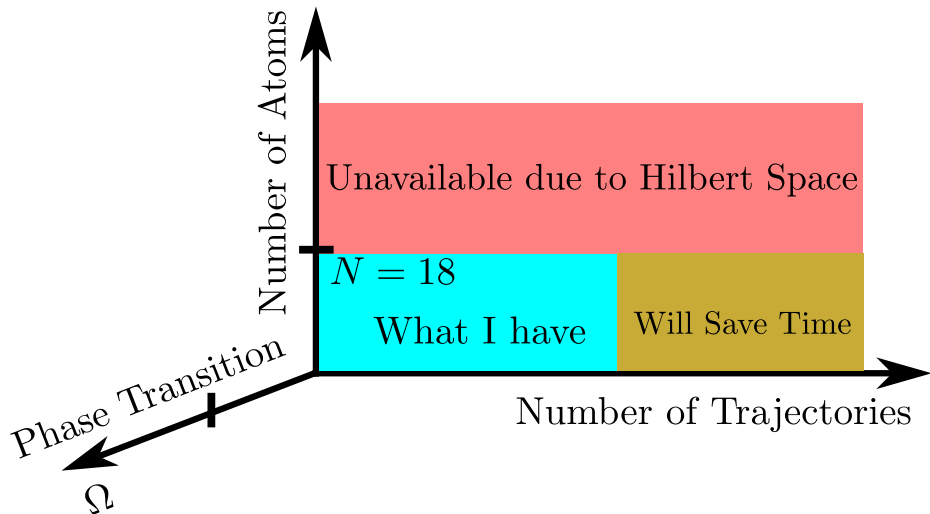


4 examples of quantum trajectories  $N = 18$  (best I can get),  $\Omega = 0.73\Omega_c$  (bistability region),  $\gamma_c = 10\gamma_s$ .

## Goal of the project

Create a model to generate trajectories similar to quantum trajectories generated by Monte Carlo wave-function simulations in QuTiP

## Why this matters?



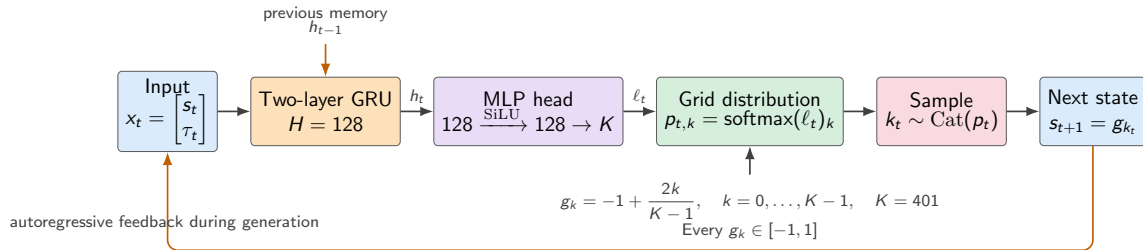
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## Dataset for fixed $N$ and $\Omega$

# Model Architecture



## Training at every time step

$$y_t = s_{t+1}^{\text{real}}, \quad q_{t,i} = 1 - \alpha, \quad q_{t,i+1} = \alpha, \quad \mathcal{L}_t = - \sum_{k=0}^{K-1} q_{t,k} \log p_{t,k}.$$

$\underbrace{\text{real } s_t \text{ as input}}_{\text{teacher forcing}} \longrightarrow \underbrace{\text{two-bin interpolated target } q_t}_{\text{continuous data on a discrete grid}} \longrightarrow \underbrace{\text{categorical cross-entropy}}_{\text{maximize next-state likelihood}}.$